Scenario C Memo (D5) -- One implicit rule across 14 / 6 / 0 crew

To: Fincantieri SDV jury

From: sdv-resource-governor team

Subject: "Nobody on the flight deck during recovery", written down and run by the engine at 14, 6 and 0 aboard

1. The rule

Today a deck officer enforces it by common sense: nobody and nothing stays on the flight deck while the drone comes in. The brief lists it as an example rule for Scenario C ("nobody on the flight deck during recovery"); Annex A.3 makes "deck clear" a flight-deck condition and the Glossary says the flight deck "Must be free of people and obstacles". We write it as: G4 may not start its recovery approach until the flight deck is verified clear, and clearing the deck is a crew task whose duration depends on how many people are aboard.

* Trigger: G4 asks to land (Scenario B min 8: battery 18 %, wind 22 kn). Annex B: "below 20 % battery: land within 12 min or ditch", so the alert at T=480 s sets a hard touchdown deadline of T=1200 s (720 s later).
* Effect: the recovery plan cannot be ready before max(ship manoeuvre done, human confirmation done, deck verified clear); touchdown follows after a 270 s approach. If that lands after the deadline, the plan is infeasible: the governor refuses it, names the binding constraint and escalates to the captain. It never plans a landing on an occupied deck.
* Crew-dependent numbers (A.6 crew response, our mid-range picks): confirmation 30 s at 14 (brief 15-60 s), 120 s at 6 (1-3 min), none at 0 (impossible uncrewed: automatic or forbidden). Clearing an occupied deck 240 s at 14 (3-5 min), 600 s at 6 (8-12 min), impossible at 0 ("uncrewed only if already clear"). G4 recovery needs 2 deck crew (ASSUMED).

2. Where the code encodes it

* src/sdv_governor/scheduling.py -- DEFAULT_TRANSITION_TIMING_POLICY (deck_clear_crew_14_s=240, deck_clear_crew_6_s=600, deck_clear_crew_0_s=None, human_confirm_crew_14_s=30, human_confirm_crew_6_s=120, human_confirm_crew_0_s=0; same values in data/ship/transition_timing_policy.yaml), compute_deck_clear_duration_s, compute_human_confirmation_duration_s.
* src/sdv_governor/arbitration.py -- _evaluate_candidate_feasibility_and_score: t_ready_s = max(maneuver_ready_s, human_confirm_done_s, deck_ready_s); witnesses flight_deck_clear_for_recovery, g4_battery_emergency_touchdown_deadline_s and g4_recovery_min_crew. arbitrate_live_conflict keeps only feasible candidates, else returns verdict NO with "the governor refuses and escalates to the captain".
* Crew level is a state input (crew_count, crew_regime, available_crew); replay.py changes it on CREW_REGIME_CHANGE events (Scenario C replay).

3. Engine results (Scenario B state, alert T=480 s, deadline T=1200 s; times after the alert)

Measured with arbitrate_live_conflict on scenario_B_arbitration_degradation.yaml, varying only crew_count / crew_regime / available_crew and deck_clear. Every decision took < 2 ms.

Crew, deck	survey_primary	rescue_primary
14, deck clear	YES: confirm 30 s, touchdown +615 s (105 s margin)	YES: touchdown +375 s
14, deck occupied	YES: deck clear 240 s, touchdown +645 s	YES: touchdown +525 s
6, deck clear	YES: confirm 120 s, touchdown +615 s	YES: touchdown +390 s (confirmation is now the slowest step)
6, deck occupied	NO: earliest touchdown +885 s > 720 s; binds time / touchdown deadline and flight_deck_clear_for_recovery; escalates	NO: same, +885 s > 720 s
0, deck clear or occupied	NO: positions / g4_recovery_min_crew binds (2 crew needed, 0 aboard); recovery refused	NO: same

4. What each crew level confirms, automates, and loses

	14 aboard	6 aboard	0 aboard
Human confirms	Bridge confirms the contested plan (30 s) and the deck team verifies the deck clear (240 s if occupied)	Same two acts, slower: 120 s confirmation, 600 s clearing	Nothing: no one can confirm
Automatic	D3 decision in < 2 s; G3 acoustic HOLD; speed/course requests; 35 kW charge reservation; D4 shedding	Same	D3, G3 HOLD and D4 shedding still run (they need no hands)
Impossible	Nothing within the deadline: clear or occupied, both priority sets land G4	Recovering onto an occupied deck: 600 s clearing misses the 12-min battery deadline	Any G4 recovery, and clearing the deck at all

5. What this means for the operator

At 14 the rule costs time but never the drone. At 6 the same rule turns the deck state at the moment of the alert into the deciding fact: keep the deck clear whenever G4 is airborne, or accept that a low-battery G4 will be lost. At 0 the governor does not pretend: it refuses the recovery, names g4_recovery_min_crew, and the cost to the loser is the drone. The jury can check this by changing only the crew level and deck state: no other code path or priority changes.